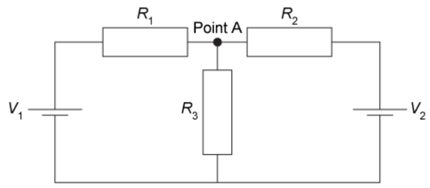
A circuit is constructed from two voltage sources (V_1 and V_2) and three resistors (R_1, R_2, and R_3).

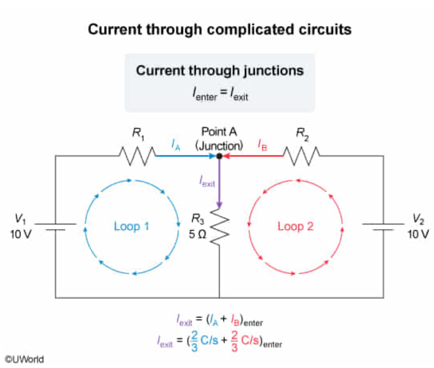


If the potential of each voltage source is 10 V and the resistance of each resistor is 5 Ω, what is the current exiting Point A into R_3?

- A. 0.7 C/s [7%]
- B. 1.0 C/s [46%]
- C. 1.3 C/s [33%]
- D. 1.6 C/s [12%]

Correct
33% answered correctly

Explanation:



In contrast to simple circuits, complicated circuits contain two or more **loops** (ie, closed pathways) sharing only some circuit components. **Junctions** are points along the circuit at which three or more circuit elements meet. According to Kirchhoff's junction rule, the sum of **current entering the junction** (I_{enter}) must be equal to the sum of **current exiting that junction** (I_{exit}):

$$I_{\text{enter}} = I_{\text{exit}}$$

Furthermore, the total voltage drop V within each loop is related to the resistance R of and current I traveling across all resistors within the loop:

$$V_{\text{n}} = I_1 R_1 + I_2 R_2 ... + I_{\text{n}} R_{\text{n}}$$

In the question, current flows through a complicated circuit with two voltage sources (V_1 and V_2) and three resistors (R_1, R_2, and R_3). The circuit contains two loops, the first of which is composed of voltage source V_1 and resistors R_1 and R_3. The second loop is composed of voltage source V_2 and resistors R_2 and R_3.

One current unique to each loop (I_A and I_B) enters Point A and another current shared by both loops exits Point A (I_{exit}):

$$I_{\text{exit}} = I_{\text{A}} + I_{\text{B}}$$

$$V_{\text{loop 1}} = I_{\text{A}} R_1 + I_{\text{exit}} R_3$$

$$V_{\text{loop 2}} = I_{\text{B}} R_2 + I_{\text{exit}} R_3$$

Since R_1 = R_2 = R_3, these variables in the equations above can be replaced with the variable R:

$$V_{\text{loop 1}} = (I_{\text{A}} + I_{\text{exit}})R$$

$$V_{\text{loop 2}} = (I_{\text{B}} + I_{\text{exit}})R$$

The equation for I_{exit} can be substituted into each of these equations, yielding:

$$V_{\text{loop 1}} = (I_A + I_A + I_B)R \qquad V_{\text{loop 2}} = (I_B + I_A + I_B)R$$
$$V_{\text{loop 1}} = (2I_A + I_B)R \qquad V_{\text{loop 2}} = (I_A + 2I_B)R$$

Substituting the values of the voltage sources and resistors into the equation for $V_{\text{loop 1}}$ and $V_{\text{loop 2}}$ gives:

$$10 = (2I_A + I_B)5 \quad 10 = (I_A + 2I_B)5$$
$$10 = 10I_A + 5I_B \quad 10 = 5I_A + 10I_B$$

Thus, there are two simultaneous equations with two unknowns. Multiplying the $V_{\text{loop 1}}$ equation by 2 gives:

$$20 = 20I_A + 10I_B$$

Subtracting the $V_{\text{loop 2}}$ equation from the equation above:

$$20 - 10 = (20I_A + 10I_B) - (5I_A + 10I_B)$$
$$10 = 20I_A - 5I_A + \cancel{10I_B} - \cancel{10I_B}$$
$$10 = 15I_A$$
$$I_A = \frac{10}{15} = \frac{2}{3}$$

Substituting this value into the equation for $V_{\text{loop 1}}$ yields:

$$10 = 10 \cdot \frac{2}{3} + 5I_B$$
$$10 = \frac{20}{3} + 5I_B$$
$$5I_B = 10 - \frac{20}{3}$$
$$5I_B = \frac{30}{3} - \frac{20}{3} = \frac{10}{3}$$
$$I_B = \frac{10}{3 \cdot 5} = \frac{2}{3}$$

From the equation for I_{exit}:

$$I_{\text{exit}} = \frac{2}{3} + \frac{2}{3}$$
$$I_{\text{exit}} = \frac{4}{3}\,\text{A} = \frac{4}{3}\,\frac{\text{C}}{\text{s}}$$

Therefore, the current exiting Point A is approximately equal to 1.3 C/s **(Choice C)**.

Educational objective:
Circuits composed of multiple loops contain junctions, points on the circuit at which three or more elements meet. The current entering any junction must be equal to the current exiting the junction.

Time spent: 0 sec
Subject: Physics

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